

Rutvik Dhopate

(857) 753-7456 | San Jose, CA - Open to Relocation | rutvikdhopate@gmail.com | [GitHub](#) | [LinkedIn](#) | [Website](#)

Summary

Data Scientist with experience deploying NLP and predictive models at Amazon and Veeco. Combined statistical modeling, ML evaluation, and data engineering to improve product information across 19 countries and reduce semiconductor manufacturing diagnostics from days to minutes.

Work Experience

Veeco Instruments Inc.

Sep 2025 — Present

Data Scientist 2

San Jose, CA

- Engineered a diagnostic application with 40 correlation-driven plots and four levels of drill-down, reducing customer escalation root-cause analysis from three days to under 30 minutes.
- Automated pre-shipment validation across 60 tools and 7,500+ parameters, achieving 100% validation coverage and closing gaps in manual quality checks.
- Achieved 97% retrieval accuracy across 2,000+ pages and 100 engineering drawings by implementing document parsing, chunking, and indexing for an air-gapped RAG pipeline.
- Delivered 2-second end-to-end latency for on-tool AI/ML applications valued at \$200,000 in annual recurring impact by building data ingestion pipelines, SQL schemas, parsers, and dashboards.

Veeco Instruments Inc

Jun 2024 — Dec 2024

Data Scientist Co-Op

San Jose, CA

- Deployed a neural network to predict wafer resistance uniformity, eliminating 12 test wafers per tool and saving 600+ wafers annually across the fleet.
- Validated a manufacturing process change through hypothesis testing, eliminating \$150,000 in supplier and shipment costs per tool.

Amazon

Jan 2022 — Aug 2023

Data Scientist

Bangalore, India

- Developed a hybrid NER-MQMRC pipeline for product information extraction across 700+ categories and 19 countries, achieving 90% precision and 70% recall and improving customer retention by 6%.
- Established benchmarking frameworks with Applied Science for image-based multimodal models, improving labeled data consistency by 10% across product classification pipelines.
- Introduced an anomaly detection pipeline to identify systematic product data errors across 800,000 products, improving precision by 4%.
- Corrected unit-of-measurement inconsistencies across two million product families using NLP models, increasing product discoverability by 4%.

Projects

Patient Eligibility Screening for Clinical Trials

Jun 2025 — Aug 2025

- Trained a PyTorch Transformer encoder for clinical trial eligibility classification, achieving 97% validation accuracy with custom DataLoaders and domain-specific BPE tokenization covering 12,000+ medical tokens.
- Published trained models and inference code to Hugging Face Hub after training on HPC NVIDIA V100 GPUs with Weights & Biases experiment tracking.

CodeSynth: XAI Agent for Statistical Modeling, Artificial Intelligence Chat-Bot

Mar 2025 — Apr 2025

- Architected a LangGraph agent to generate and evaluate statistical modeling code, achieving 88% task success on DS1000 benchmark problems through prompt optimization and LLM-as-a-Judge evaluation.
- Created a sandboxed Python runtime to execute generated code and capture evaluation outputs, reducing manual workload by 80%.

Skills

- Languages, ML & AI:** Python, SQL, R, PyTorch, Scikit-Learn, LangChain, Transformers, RAG, LLM Evaluation, Causal Inference
- Data & Statistics:** Snowflake, AWS, Kafka, FastAPI, Weights & Biases, A/B Testing, Hypothesis Testing, BLEU, ROUGE, Bayesian Modeling

Education

Northeastern University

Sep 2023 — Aug 2025

Master of Science, Data Science (GPA: 4.00 / 4.00)

Boston, MA

NMIMS University

Jul 2020 — May 2022

Master of Science, Applied Statistics and Analytics (GPA: 3.73 / 4.00)

Mumbai, India